

SAMYA DUTTA

☎ +91 84200 30435 ✉ samyadutta29@gmail.com 🔗 [linkedin.com/in/samya-dutta](https://www.linkedin.com/in/samya-dutta) 🐙 github.com/SamyaDutta

Education

Institute of Engineering and Management, Salt Lake, Kolkata 2022-2026
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) **DGPA: 9.58**

Experience

GreyB | *Research Analyst Intern* Jan 2026 - June 2026

- Conducted in-depth prior art searches and technical analyses as a Research Analyst at GreyB to evaluate patentability, identify infringement risks, and support IP strategy decisions.

IEEE EMBS PUNE SECTION | *Research Intern* June 2025 – July 2025

- Trained 13 ML models for epilepsy classification using Electronic Health Records Biomedical Dataset for resource-scarce areas, achieving 91% precision and 0.92 F1 score.

IEM RESEARCH FOUNDATION | *Generative AI Intern* May 2025 – July 2025

- Developed a multi-agent HR automation system using LangGraph, LangChain, RAG, ChromaDB, Sentence Transformers, Pydantic, and FastAPI to automate resume screening, candidate shortlisting, interview scheduling, employee churn analysis, interview transcript analysis, propensity analysis, grade fitment, and ROI/learning analysis.

Projects

Logistics AI-Powered Supply Chain Decision Intelligence-System Sept 2026

- Developed an AI-powered decision-intelligence system that enables logistics teams to identify high-risk shipments, temperature anomalies, and delay risks by combining fleet telemetry, live weather conditions, and cold-chain SOPs; built a LangGraph agent with 3 specialized tools for read-only SQL telemetry analysis, live corridor-weather retrieval, and Pinecone-based SOP/RAG retrieval, supporting multi-tool reasoning in a single query and analysis of the top 5/10 highest-risk fleet records; implemented SQL semantic views, session-aware conversations, BGE-M3 embeddings, audit logging, and administrator-level audit review using LangChain, Groq, SQL Server, Pinecone, Open-Meteo, and Streamlit.

TripPilot AI — Multi-Agent Travel Planning System Sept 2026

- Built a multi-agent AI travel planning system using LangGraph, LangChain, Groq, MCP, FastAPI, and PostgreSQL, orchestrating 5 specialized agents for flight, hotel, weather, budget, and itinerary planning through a Supervisor-driven workflow; integrated Tavily, AviationStack, and custom Weather MCP tools for real-time travel data, implemented LLM-based input guardrails, human-in-the-Loop approval with revision feedback, conditional agent routing, and PostgreSQL checkpointing for persistent multi-turn travel sessions, exposed through 3 FastAPI endpoints for planning, approval, and health monitoring.

Recruvia — Recruitment Automation System April 2025

- Developed Recruvia, an AI-powered multi-agent recruitment automation system using CrewAI, Groq LLMs, ChromaDB, Streamlit, Pandas, and Gmail SMTP, orchestrating 5 specialized agents for natural-language HR query interpretation, candidate profile retrieval, CV screening, recruitment reporting, and interview scheduling; indexed candidate profiles using 8 attributes (name, role, location, skills, experience, achievements, education, certifications) in a persistent ChromaDB vector store, retrieved and ranked the top 5 relevant candidates through similarity search, evaluated candidates using a 100-point scoring framework (40 points experience + 60 points skill match), generated recruitment reports with recommendations and next steps, and automated personalized interview invitations with Google Meet links.

Skills

Programming & Scripting: Java, Python, SQL

RAG & Agentic AI Systems: Retrieval-Augmented Generation (RAG), Agentic AI Workflows, Multi-Agent Systems

Frameworks & Tools: LangChain, LangGraph, CrewAI

Vector Databases & Retrieval: ChromaDB, Pinecone

Relevant Coursework: Data Structures and Algorithms, Operating System, Object Oriented Programming, Database Management System, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs)

Leadership / Extracurricular

IEEE Computer Society, IEM Student Branch Chapter

- Chairperson Feb 2025 - Mar 2026
- Secretary Feb 2024 - Jan 2025

Achievements

GATE 2025 and GATE 2026 Qualified

Director's Award for Best Scientific Mind in IEM Kolkata 2026

Best IEEE Computer Society SBC award across India in AICSSYC 2025 as Chairperson

Best IEEE SBC Award in Kolkata Section 2025 as Chairperson

IEEE India Council Hackathon Finalist 2023